Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
1   #include<stdio.h>
2   void func1(int *arr){
3     for(int i=1; i<=3;i++)
4       (i-1)[arr] = (i%3);
5   }
6   void func2 (int a[1] , int arr){
7     a[0] = arr + 1;
8   }
9  int main(){
10   int arr[4] = {0,0,0,0};
11   func1(arr+1);
12   for(int i=0; i<4; i++)
13         printf("%d ", i[arr] );
14   printf("\n");
15   func2(arr, *arr);
16   for(int i=0; i<4; i++)
17         printf("%d ", i[arr] );
18   return 0;
19 }
```

Output
0 1 2 0
1 1 2 0

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
 int temp = arr + 1;
 a = &temp;
}
```

Explain your answer

No, As now, in the func2( ), a is pointing to the tmp, and Any change is reflect to the tmp.

Q. 2: Write the output of the following sequence of instructions **long arr2[100]; int arr1[100], a1=(&arr2[50]), a2=(&arr2[10]), b1=(&arr1[50]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

1280 1280

As, in the Arr. store in memory as continous memory

32
40
1280